Machine Learning for Breast Cancer Treatment Response Prediction Using Clinical and MRI Radiomics Features

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***Abstract*—This study presents an end-to-end machine learning pipeline for predicting breast cancer treatment response using clinical and MRI-derived radiomics features. Two tasks were investigated: pathological complete response (pCR) classification and relapse-free survival (RFS) regression. The dataset contained 400 patients with 11 clinical features and 107 MRI-derived radiomics features. Missing values encoded as 999 were replaced and imputed, followed by feature scaling and feature selection. Clinically important features, including ER, HER2 and Gene, were retained throughout the modelling process. For pCR classification, Logistic Regression, Random Forest, Gradient Boosting, Extra Trees and Support Vector Machine models were compared. The best tuned pCR model was an RBF-kernel Support Vector Machine with 25 selected features, achieving a cross-validated balanced accuracy of 0.6608. For RFS regression, the best tuned model was a Random Forest Regressor with 20 selected features, achieving a cross-validated MAE of 20.6832. The results suggest that pCR prediction is moderately feasible, while RFS prediction remains challenging using pre-treatment clinical and radiomics features alone.**

Keywords—breast cancer, radiomics, machine learning, pathological complete response, relapse-free survival, feature selection, support vector machine, random forest.

I. Introduction

Breast cancer is the most common cancer among women in the UK, with around 59,000 new female cases each year [11]. Neoadjuvant chemotherapy is commonly used before surgery to reduce the size of locally advanced tumours and improve treatment planning. However, chemotherapy is associated with systemic toxicity and is not equally effective for all patients. Complete tumour resolution at surgery, known as pathological complete response (pCR), is associated with improved long-term outcomes, including better relapse-free survival [1]. RFS refers to the length of time after primary treatment during which a patient remains alive without signs or symptoms of relapse. However, only a minority of patients receiving chemotherapy achieve pCR, while many have residual disease and variable prognoses. Therefore, predicting pCR and RFS using information available before chemotherapy could support better patient stratification and more personalised treatment planning.

In addition to clinical biomarkers, medical imaging contains quantitative information that may help describe tumour phenotype and heterogeneity. Radiomics refers to the extraction of numerical features from medical images, including shape, intensity and texture descriptors [2], [3]. When extracted from pre-treatment magnetic resonance imaging scans, radiomics features may provide additional information about tumour characteristics that are not fully captured by conventional clinical measurements. Previous work using the I-SPY 2 trial has shown that multi-feature breast MRI can support prediction of treatment response to neoadjuvant therapy [1].

This project investigates the use of machine learning models to predict two treatment-related outcomes in breast cancer patients. The first task is pCR classification, where the aim is to predict whether a patient achieves pathological complete response. The second task is RFS regression, where the aim is to estimate relapse-free survival time. These tasks represent different prediction challenges: pCR is a binary classification problem with class imbalance, while RFS is a continuous regression problem involving long-term clinical outcome estimation.

The main contribution of this work is the development of a reproducible machine learning pipeline covering data preprocessing, missing value handling, feature scaling, feature selection, model comparison, hyperparameter tuning and final prediction generation. Multiple supervised learning models were evaluated for both tasks. For pCR classification, balanced accuracy was used as the primary metric to account for class imbalance. For RFS regression, mean absolute error was used as the main evaluation metric because it directly measures prediction error in the original survival-time scale.

This report presents the dataset preparation process, model development approach, experimental results and limitations of the proposed pipeline. The goal is not to claim clinical deployment readiness, but to provide a structured and interpretable machine learning case study for treatment response prediction using clinical and MRI-derived radiomics data.

II. Literature Review

A. MRI Radiomics for Breast Cancer Response Prediction

Radiomics has become an important approach in medical image analysis because it converts medical images into quantitative features that describe tumour shape, intensity, texture and heterogeneity [2], [3]. In breast cancer, MRI-derived radiomics features have been investigated for predicting response to neoadjuvant chemotherapy before treatment is completed [1], [4]. This is useful because early prediction of treatment response may support more personalised treatment planning and reduce unnecessary exposure to ineffective therapy.

Previous studies have shown that radiomics features extracted from pre-treatment MRI can contribute useful information for predicting pathological complete response. Clinical biomarkers such as ER, HER2 and gene-related variables are also commonly used because they reflect important biological characteristics of breast cancer. Combining clinical variables with imaging-derived radiomics features is therefore often considered more informative than using either source individually [1], [2].

B. Machine Learning Models for Radiomics-Based Prediction

Machine learning is commonly used in radiomics because the extracted feature space is often high-dimensional and may contain complex relationships with clinical outcomes. Linear models such as Logistic Regression are widely used because they are interpretable and can perform well when the selected features have approximately linear relationships with the outcome. This makes them useful as baseline models in medical prediction tasks.

Non-linear models such as Support Vector Machines, Random Forests and Gradient Boosting can capture more complex feature interactions. Support Vector Machines are suitable for smaller datasets with high-dimensional feature spaces because they construct decision boundaries based on support vectors and margins [7]. Random Forests are ensemble tree-based models that aggregate multiple decision trees and can model non-linear feature interactions [8]. However, more complex models may also overfit when the dataset is small relative to the number of features. For this reason, model comparison and cross-validation are necessary to identify which method generalises best.

Deep learning methods have also been applied to breast MRI analysis, particularly when raw imaging data are available. However, deep models usually require larger datasets and more computational resources. In limited-sample radiomics studies, classical machine learning models may be more appropriate because they are easier to train, easier to interpret and less likely to overfit severely.

C. *Relapse-Free Survival Prediction*

Compared with pCR classification, relapse-free survival prediction is more challenging because RFS is a long-term outcome influenced by many biological, treatment-related and follow-up factors. While pCR reflects immediate treatment response, RFS depends on whether cancer signs or symptoms return over time. As a result, pre-treatment clinical and radiomics features alone may not fully explain survival variation.

Previous radiomics studies have explored the relationship between MRI features and recurrence or relapse-related endpoints. These studies suggest that imaging features may contain prognostic information, but survival prediction often remains difficult due to limited sample sizes, censored or noisy outcome data and the complexity of disease progression. Therefore, regression models for RFS should be interpreted carefully, and performance should be assessed using error-based metrics such as mean absolute error rather than relying only on explained variance.

D. *Methodological Challenges in Radiomics Modelling*

Radiomics-based machine learning faces several methodological challenges. First, the number of radiomics features can be large compared with the number of patients, increasing the risk of overfitting. Feature selection is therefore important to reduce dimensionality, remove less informative predictors and improve model stability [9]. However, clinically important features should not be removed automatically, especially when prior knowledge suggests that they are relevant to the outcome.

Second, class imbalance is common in pCR prediction because the number of complete responders is usually smaller than the number of non-responders. In such cases, standard accuracy can be misleading because a model may appear accurate while mainly predicting the majority class. Balanced accuracy, precision, recall and confusion matrices are more informative for assessing classification performance under imbalance [5].

Third, radiomics features may be sensitive to MRI acquisition settings, segmentation differences and preprocessing choices. This can reduce reproducibility across datasets or institutions. Robust preprocessing, cross-validation and cautious interpretation are therefore essential. Standardisation initiatives have highlighted the importance of consistent radiomics feature extraction for improving reproducibility [10]. External validation would be needed before any model could be considered clinically reliable.

E. *Positioning of This Work*

This project builds on previous radiomics-based treatment response studies by implementing a complete and reproducible machine learning pipeline for two related tasks: pCR classification and RFS regression. The work compares multiple supervised learning models, applies missing value handling and feature scaling, performs mandatory feature selection while retaining ER, HER2 and Gene, and evaluates model performance using task-appropriate metrics. The aim is to provide a structured machine learning case study rather than a clinically deployable prediction system.

III. METHODOLOGY

A. *Dataset Description*

The dataset used in this project consists of 400 breast cancer patient records. Each patient sample contains 11 clinical features and 107 MRI-derived radiomics features. The clinical features include patient and tumour-related biomarkers such as Age, ER, PgR, HER2, Triple Negative status, tumour stage and gene-related information. The radiomics features were extracted from tumour regions in MRI scans and represent quantitative imaging descriptors such as first-order, shape and texture-based measurements.

Two outcome variables were considered. The first was pathological complete response, denoted as pCR, which was treated as a binary classification target. A value of 1 indicates complete response, while 0 indicates residual disease. The second outcome was relapse-free survival, denoted as RFS, which was treated as a continuous regression target. The final test dataset contained patient IDs and feature values only, without outcome labels, and was used only for prediction generation.

B. *Data Preprocessing*

The raw dataset used the placeholder value 999 to represent missing values. Therefore, the first preprocessing step replaced all 999 values with missing values. For the pCR classification task, samples with missing pCR labels were removed because they could not be used for supervised classification. The RFS regression task retained all samples with available RFS target values.

After separating the input features from the target variables, missing feature values were imputed using median imputation. Median imputation was selected because it is more robust to extreme values than mean imputation, which is useful for radiomics features that may contain skewed distributions or outliers. Feature scaling was then performed using standardisation so that each feature had a mean of zero and unit variance. This was particularly important for distance- and margin-based models such as Support Vector Machines.

The preprocessing steps were implemented using Scikit-learn pipelines [6]. This ensured that imputation, scaling, feature selection and modelling were fitted only on the training folds during cross-validation, reducing the risk of data leakage.

C. *Feature Selection*

The dataset contains a relatively high number of features compared with the number of patient samples. This increases the risk of overfitting and reduces model stability. Feature selection was therefore applied to reduce dimensionality and retain only the most informative predictors [9].

A custom feature selector was implemented using a univariate statistical scoring approach. For the pCR classification task, ANOVA F-statistics were used to rank features based on their relationship with the binary class label. For the RFS regression task, F-regression scores were used to rank features based on their relationship with the continuous survival target.

Domain knowledge was also incorporated into the feature selection process. The clinical features ER, HER2 and Gene were treated as mandatory predictors and were retained in the modelling process regardless of their univariate ranking. This was done because these variables are clinically relevant to breast cancer treatment response and were considered important prior features. The final tuned pCR model used 25 selected features, while the final tuned RFS model used 20 selected features.

D. *pCR Classification Models*

The pCR prediction task was formulated as a binary classification problem. Several supervised machine learning models were compared:

- Logistic Regression
- Random Forest Classifier
- Gradient Boosting Classifier
- Extra Trees Classifier
- Support Vector Machine

These models were selected to compare both linear and non-linear learning approaches. Logistic Regression was included as an interpretable linear baseline. Random Forest, Gradient Boosting and Extra Trees were included as tree-based ensemble methods capable of modelling non-linear feature interactions [8]. Support Vector Machine was included because it is suitable for smaller datasets with high-dimensional feature spaces [7].

Initial model comparison was performed using 5-fold stratified cross-validation and a separate validation split. Stratification was used to preserve the proportion of pCR and non-pCR samples across folds because the classification target was imbalanced. Balanced accuracy was used as the primary evaluation metric because standard accuracy can be misleading when one class is more common than the other.

After comparison, the strongest pCR candidates were tuned using GridSearchCV. The final selected pCR model was an RBF-kernel Support Vector Machine with mandatory feature selection. The best hyperparameters were C = 0.3, gamma = 0.01 and 25 selected features.

E. *RFS Regression Models*

The RFS prediction task was formulated as a regression problem. Several regression models were evaluated:

- Ridge Regression
- Random Forest Regressor
- Gradient Boosting Regressor
- Support Vector Regressor

These models were selected to compare linear, kernel-based and tree-based regression approaches. Ridge Regression was used as a regularised linear baseline. Support Vector Regression was included to model non-linear relationships using an RBF kernel. Random Forest and Gradient Boosting regressors were included because ensemble tree methods can capture non-linear interactions between clinical and radiomics features.

The models were evaluated using 5-fold cross-validation. Mean Absolute Error was used as the main evaluation metric because it measures the average prediction error in the original RFS unit and is easier to interpret than squared-error metrics. Root Mean Squared Error and R^2 were also calculated as secondary metrics to provide additional insight into prediction error magnitude and explained variance.

Hyperparameter tuning was performed using GridSearchCV. The final selected RFS model was a Random Forest Regressor with mandatory feature selection. The best configuration used 20 selected features, 200 estimators and a minimum leaf size of 3.

F. *Hyperparameter Tuning*

Hyperparameter tuning was performed separately for the classification and regression tasks. For pCR classification, the grid search compared Logistic Regression and Support Vector Machine configurations with different feature counts and regularisation parameters. The grid included different values of C, gamma and number of selected features. The best model was selected based on cross-validated balanced accuracy.

For RFS regression, the grid search compared Support Vector Regression, Random Forest Regression and Gradient Boosting Regression using different feature counts and model-specific hyperparameters. Random Forest parameters included the number of estimators, maximum tree depth and minimum samples per leaf. The best model was selected based on the lowest cross-validated mean absolute error.

G. *Evaluation Strategy*

For pCR classification, the primary metric was balanced accuracy, defined as the average of sensitivity and specificity. This metric was prioritised because the pCR classes were imbalanced. Precision, recall, F1-score and confusion matrices were also used to analyse class-specific performance.

For RFS regression, the primary metric was mean absolute error. This metric was selected because it directly measures the average absolute difference between the predicted and actual RFS values. RMSE and R^2 were reported as secondary metrics. RMSE penalises larger errors more heavily, while R^2 provides an indication of how much variance in the target variable is explained by the model.

The final test dataset was used only to generate prediction files. Since the true pCR and RFS labels were not available in the final test file, final test performance could not be directly calculated. Therefore, model selection and performance reporting were based on cross-validation results from the labelled training dataset.

IV. Results and Discussion

A. *pCR Classification Results*

The pCR classification task was evaluated using balanced accuracy as the primary metric because the dataset was imbalanced, with fewer pCR-positive cases than non-pCR cases. Table 1 shows the initial model comparison before final hyperparameter tuning and feature selection.

TABLE I. BASELINE pCR MODEL COMPARISON

Model	*CV Balanced Accuracy*	*Validation Balanced Accuracy*
Support Vector Machine	0.6760 ± 0.0558	0.5930
Logistic Regression	0.6643 ± 0.0906	0.6063
Extra Trees	0.6253 ± 0.0733	0.5527
Random Forest	0.5693 ± 0.0381	0.5560
Gradient Boosting	0.5501 ± 0.0417	0.5398

Among the baseline models, Support Vector Machine achieved the highest cross-validated balanced accuracy of 0.6760, while Logistic Regression achieved the highest validation balanced accuracy of 0.6063. Random Forest, Extra Trees and Gradient Boosting produced lower balanced accuracy scores, suggesting that tree-based ensemble models were less effective for this classification task under the tested settings.

The strongest models were then further tuned using GridSearchCV with mandatory feature selection. The final selected pCR model was an RBF-kernel Support Vector Machine using 25 selected features. The mandatory clinical features ER, HER2 and Gene were retained in the selected feature set.

TABLE II. FINAL TUNED pCR MODEL

Model	*Selected Features*	*Best Parameters*	*CV Balanced Accuracy*
SVM + Feature Selection	25	C = 0.3, gamma = 0.01	0.6608

Although the tuned SVM had a slightly lower cross-validated balanced accuracy than the untuned SVM comparison, it was selected as the final model because it combined hyperparameter tuning, feature selection and mandatory clinical feature retention in a single reproducible pipeline. This made the final model more controlled and less likely to rely on noisy or redundant radiomics features.

The classification results indicate moderate predictive ability. The model achieved performance above random-level balanced accuracy, but the balanced accuracy remained far from clinical-level reliability. This suggests that the available clinical and MRI-derived radiomics features contain useful information for pCR prediction, but the small dataset size and class imbalance limit performance.

B. RFS Regression Results

The RFS regression task was evaluated using mean absolute error as the primary metric. MAE was selected because it measures the average absolute prediction error in the original RFS scale. RMSE and R^2 were also calculated as secondary metrics.

TABLE III. BASELINE RFS MODEL COMPARISON

Model	*CV MAE*	*Validation MAE*	*Validation RMSE*	*Validation R^2*
Random Forest	20.8671 ± 1.3072	22.5611	28.8335	-0.0417
SVR	21.6875 ± 1.6687	21.7850	28.5136	-0.0188
Gradient Boosting	22.0539 ± 1.3494	22.6772	29.1627	-0.0657
Ridge Regression	35.6503 ± 24.6424	23.5586	30.0126	-0.1287

The baseline comparison showed that Random Forest achieved the lowest cross-validated MAE, while SVR achieved the lowest validation MAE on the single holdout split. Since cross-validation provides a more reliable estimate across multiple folds, Random Forest was selected for further tuning.

After feature selection and hyperparameter tuning, the final selected RFS model was a Random Forest Regressor using 20 selected features. As with the pCR task, ER, HER2 and Gene were retained during feature selection.

TABLE IV. FINAL TUNED RFS MODEL

Model	*Selected Features*	*Best Parameters*	*CV MAE*
Random Forest Regressor + Feature Selection	20	n_estimators = 200, min_samples_leaf = 3	20.6832

The tuned Random Forest Regressor achieved a cross-validated MAE of 20.6832. This was a slight improvement over the baseline Random Forest MAE of 20.8671. However, the R^2 values from the baseline experiments were close to zero or negative, suggesting that the models explained only a limited amount of variance in RFS outcomes.

These results show that RFS prediction was more difficult than pCR classification. This is expected because relapse-free survival is a long-term outcome that may depend on additional biological, treatment-related and follow-up factors that are not fully represented by pre-treatment imaging and clinical features.

C. Final Test Prediction Generation

After training and tuning the final models, prediction scripts were used to generate outputs for the final test dataset. The final test file contained 133 patients and did not include the true pCR or RFS labels. Therefore, final test accuracy and MAE could not be calculated directly.

The final prediction scripts generated two output files:

- pcr_prediction.csv
- rfs_prediction.csv

Each file contained 133 rows, with the first column containing the patient ID and the second column containing the predicted outcome. The pCR output contained binary class predictions, while the RFS output contained continuous predicted survival values. This confirmed that the final prediction pipeline could run successfully on unseen test data with the same feature structure.

D. Discussion

The results show that pCR classification was more successful than RFS regression. The final pCR model achieved a cross-validated balanced accuracy of 0.6608, indicating moderate predictive performance. This suggests that the combination of clinical biomarkers and MRI-derived radiomics features contains useful signal for identifying treatment response. However, the performance is still limited and should not be interpreted as clinically reliable.

The Support Vector Machine was selected as the final pCR classifier because it performed well in cross-validation and is suitable for smaller, high-dimensional datasets. The use of an RBF kernel allowed the model to capture non-linear relationships between features and treatment response. Feature selection also helped reduce dimensionality, which is important because the dataset contained 118 input features but only 400 patient samples. Retaining ER, HER2 and Gene ensured that clinically important predictors remained available to the model.

The baseline comparison showed that Logistic Regression and SVM performed better than the tree-based models for pCR classification. This may be because the selected features had relationships that could be captured effectively by margin-based or linear-style decision boundaries. In contrast, Random Forest, Extra Trees and Gradient Boosting may have been more affected by the small dataset size, class imbalance and high-dimensional radiomics feature space. More complex models can overfit when the number of training samples is limited.

RFS prediction was more challenging. Although the tuned Random Forest Regressor achieved the lowest cross-validated MAE of 20.6832, the improvement over the baseline models was small. The low and negative R^2 values observed in the regression experiments suggest that the models explained little variance in relapse-free survival. This indicates that RFS may not be strongly predictable from the available pre-treatment clinical and radiomics features alone.

There are several possible reasons for the weaker RFS performance. First, relapse-free survival is a long-term clinical outcome and may depend on factors not included in the dataset, such as treatment details, follow-up information, genetic markers, lifestyle factors or post-treatment imaging. Second, RFS values may contain greater noise and variability than pCR labels. Third, the dataset size is relatively small for high-dimensional radiomics modelling, making it difficult for regression models to learn stable survival patterns.

The results also highlight the importance of using appropriate evaluation metrics. For pCR classification, normal accuracy can be misleading because the classes are imbalanced. Balanced accuracy was therefore more suitable because it accounts for both sensitivity and specificity. For RFS regression, MAE was prioritised because it measures prediction error in the original target scale and is easier to interpret than squared-error metrics.

Overall, the project demonstrates a complete machine learning workflow for healthcare prediction, including preprocessing, imputation, scaling, feature selection, model comparison, hyperparameter tuning and final prediction generation. The pCR model showed moderate performance, while RFS prediction remained limited. This supports the conclusion that radiomics and clinical features may contribute to treatment response prediction, but stronger datasets, external validation and survival-specific modelling would be required before such models could be considered clinically useful.

V. Limitations, Future Work and Reflection

A. Limitations

This project has several limitations. First, the dataset size was relatively small compared with the number of input features. The training dataset contained 400 patient records but 118 input features, including 107 MRI-derived radiomics features. This high feature-to-sample ratio increases the risk of overfitting and may reduce the stability of the selected features and trained models.

Second, the RFS regression task showed limited predictive strength. Although the final Random Forest Regressor achieved the lowest cross-validated MAE, the regression models produced low or negative R^2 values during evaluation. This suggests that the available pre-treatment clinical and radiomics features were not sufficient to explain much of the variation in relapse-free survival.

Third, the project did not include external validation using an independent dataset from another institution or imaging source. As a result, the generalisability of the trained models cannot be fully assessed. Radiomics features may be sensitive to imaging acquisition settings, segmentation differences and preprocessing procedures, so performance may change when tested on data from different scanners or clinical centres.

Fourth, the final test dataset did not contain true pCR or RFS labels. Therefore, the final prediction scripts could be tested for successful execution and correct output formatting, but final test performance could not be directly calculated. Model selection and evaluation were therefore based on cross-validation results from the labelled training dataset.

Finally, the models used in this work are not intended for clinical decision-making. The project demonstrates a machine learning workflow, but further validation, clinical review and prospective testing would be required before any similar system could be considered for healthcare use.

B. Future Work

Future work could improve the project in several ways. First, larger and more diverse datasets should be used to improve model generalisation. Additional patient records from different imaging centres would help reduce overfitting and allow the models to be tested across different clinical settings.

Second, external validation should be performed using a fully independent test dataset with known pCR and RFS outcomes. This would provide a more reliable estimate of real-world performance and help identify whether the models generalise beyond the training data.

Third, the feature engineering process could be improved by comparing clinical-only, radiomics-only and combined clinical-radiomics models. This would make it clearer how much predictive value is added by MRI-derived radiomics features compared with standard clinical biomarkers.

Fourth, more robust preprocessing methods could be explored, such as outlier handling, robust scaling and radiomics harmonisation techniques. These methods may reduce the effect of unstable imaging features and improve reproducibility.

Fifth, survival-specific modelling techniques could be considered for RFS prediction. Instead of treating RFS purely as a standard regression task, future work could explore survival analysis models that are designed for time-to-event data.

Finally, model interpretability could be improved by analysing selected features and feature importance. This would help explain which clinical or radiomics features contributed most to the predictions and would make the results more useful for discussion in a healthcare context.

C. Reflection

This project strengthened my understanding of how machine learning pipelines are developed for real-world healthcare data. The work showed that model performance depends not only on the choice of algorithm, but also on preprocessing, missing value handling, feature selection, class imbalance and evaluation strategy.

One important lesson was that higher model complexity does not always lead to better performance. For pCR classification, simpler or margin-based models such as Logistic Regression and SVM performed better than several tree-based ensemble models. This suggests that careful model comparison is more important than assuming that a more complex algorithm will automatically produce better results.

Another key reflection is that healthcare prediction tasks are difficult and should be interpreted cautiously. The pCR model achieved moderate performance, but the RFS regression task remained weak. This highlighted the importance of reporting limitations honestly rather than overclaiming model effectiveness.

The project also improved my ability to build reproducible machine learning workflows using Scikit-learn pipelines. By combining preprocessing, feature selection and modelling in a single pipeline, the implementation reduced the risk of data leakage and made the workflow easier to reuse on unseen test data.

Overall, this project provided valuable experience in applying machine learning to clinical and radiomics data. It demonstrated the importance of rigorous evaluation, realistic interpretation and responsible communication when working on healthcare-related AI projects.

VI. Conclusion

This project developed a complete machine learning pipeline for predicting breast cancer treatment response using clinical and MRI-derived radiomics features. For pCR classification, the best final model was an RBF-kernel Support Vector Machine with mandatory feature selection, achieving a cross-validated balanced accuracy of 0.6608. For RFS regression, the best final model was a Random Forest Regressor with mandatory feature selection, achieving a cross-validated MAE of 20.6832.

The results show that pCR classification was moderately feasible, while RFS regression remained more challenging. This suggests that pre-treatment clinical and radiomics features contain some predictive signal for treatment response but are not sufficient for strong long-term survival prediction. Overall, the work demonstrates a reproducible healthcare machine learning workflow, while also highlighting the need for larger datasets, external validation and survival-specific modelling before clinical use.

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